

Safety & Emergency Protocol Manual

General Office Safety Plan

This manual implements instructions & procedures for employees to prevent accidents.

1. **Safe Workplace Conditions:**

All work areas, aisles and hallways shall be kept clear at all times.

All exits are accessible and clearly marked and properly illuminated.

All work areas, storage are kept in a sanitary condition, floors shall be clean and, so far as possible, a dry condition.

Enclosures or guards are used in locations where electric equipment could be exposed to physical damage.

Electric equipment is appropriately connected and grounded.

All waste is stored in receptacles that are in good condition and clean.

All waste and garbage is removed in the proper manner.

Store materials in a stable manner, so that it will not be subject to falling.

Keep walk-ways and work areas free of electrical cords.

Never make repairs to light fixtures or change light bulbs unless authorized to do so by Dr. Leong.

2. **Safe Work Practices.**

Do not lift by yourself equipment and materials over 20 lbs. When carrying loads, care should be exercised to avoid overexertion and strain. Use proper lifting and reaching techniques.

Use adjustable chairs in the operatory in order to reduce musculoskeletal injuries.

Use appropriate safety eyewear/goggles when working with UV light or laser.

Employees must exercise caution in moving about the office.

Employees are trained in the proper use of autoclaves and proper ventilation is provided whenever utilizing autoclaves.

Employees that must use repetitive motions are instructed about proper techniques to avoid repetitive motion injuries.

Employees must report all unsafe conditions and symptoms of injury to Dr. Leong.

3. **Protective Equipment.**

- Always wear protective equipment when cleaning operatories, labs and or equipment, treating patients and dental assisting duties.
- Eyewash station is available to all employees who have been instructed in its proper use.

Fire and Emergency Action Plan

This plan sets out instructions and procedures for employees to follow in the event of fire or emergency evacuation at the dental practice.

1. Fire & Emergency

- In the event of fire, call local fire department (911), notify other employees, patients and visitors, and exit the building through the main entrance doors (there is only one exit in the office). All employees at the practice at time of evacuation shall meet outside the entrance to the main building lobby to assure others of their safe exit.
- In the event of an earthquake, move safely to areas in the facility that offer protection from or are away from falling items. Employees also may direct patients and visitors to these areas.
- In the event of a power outage and the practice relies on electric illumination, wait until eyes adjust to the low light prior to moving.

2. Fire Prevention

When utilizing heat producing equipment, make sure that the area is clear of all fire hazards and all sources of potential fires are eliminated.

Have fire extinguishers available at all times when utilizing heat-producing equipment.

Fire extinguisher is located under the front desk.

Know the location of fire fighting equipment in the work area and have knowledge of its use and application. Use these devices only in cases of fire.

Keep all flammables away from ignition sources.

Maintain sufficient access and working space around electric equipment.

3. Exits

All exit doors and or openings must be clear and unobstructed at all times.

All exits are arranged so that it will not be necessary to travel toward any area with hazard in order to reach the nearest emergency and evacuation route.

Aisles and hallways shall be kept clear at all times.

4. Compressed Gas Cylinders (Oxygen and Nitrous)

- All gas cylinders shall have their contents clearly marked on the outside of each cylinder.
- Cylinders must be placed and secured in an upright position, including storage and transfer.
- Cylinder valves must be protected with caps and guards when not in use.

- All leaking or defective cylinders must be removed from service promptly, tagged as inoperable and placed in an open space removed from work area.
- All operators are required to inspect equipment prior to utilization.
- 4. Regulated waste**
 - Amalgam waste shall be stored in the amalgam separator container and disposed off once full in the manner described on the packaging.
 - Sharps such as blades, needles and broken glass shall be placed in the sharps container and disposed off once full in the manner described on the packaging.
- 5. Training and Education in Fire and Emergency Safety**
 - All employees receive education on precautionary measures for fire and emergency as stated above.
 - All employees are trained on how to make a safe and orderly exit from the facility.

Exposure control plan

Minimizing potential occupational exposures to infectious microorganisms is the primary objective of this dental office's Exposure Control Plan.

Job classifications

List of job classifications where all employees have exposure potential: dental assistant, dental hygienist, dentist, sterilization technician

Exposure Incidents

Each exposure incident must be reported to Dr. Leong. Delays in reporting exposures can reduce the effectiveness of post exposure treatments, so it is critical to report incidents quickly and seek prompt treatment.

An "exposure incident" is a specific eye, mouth, or other mucous membrane, non-intact skin, or parenteral contact with blood or other potentially infectious materials as a result of performing a work duty.

Sharps Injury Log

Each exposure incident that includes a sharp must be recorded. A "sharp" is any dental instrument or object that may penetrate the skin or any other part of the body, including but not limited to needles, burs, instruments, blades, wires and broken glass.

Procedures

Tasks and procedures that have occupational exposure to blood and other potentially infectious material

- A. Clinical treatment procedures include preventative, periodontal, restorative, surgical and orthodontic.
- B. Radiographic procedures
- C. Sterilization procedures

D. Treatment room cleaning/disinfecting

Methods of Compliance

- *Standard Precautions and Infection Control Procedures*

This dental office utilizes “standard precautions” as one approach to infection control. Standard precautions integrate and expand the elements of universal precautions into a standard of care designed to protect dental healthcare professionals and patients from pathogens that can be spread by blood or any other body fluid, excretion, or secretion. Standard precautions apply to contact with 1) blood; 2) all body fluids, secretions, and excretions (except sweat), regardless of whether they contain blood; 3) non-intact skin; and 4) mucous membranes. Other effective infection control procedures include a combination of engineering and work practice controls.

This office also complies with Dental Board of California infection control regulations, some of which overlap and are more stringent than Cal/OSHA requirements.

- *Engineering and Work Practice Controls*

Engineering controls are controls, such as heat sterilizers, chemical disinfectants, instrument washers, sharps disposal containers, ultrasonic cleaners, high velocity evacuation, needle recapping devices, needleless systems, sharps with engineered sharps injury protection, etc., that isolate or remove the bloodborne pathogen hazard from the workplace. Work practice controls are controls that reduce the likelihood of exposure by defining the manner in which a task is performed (e.g., prohibiting recapping of needles by a two-handed technique and use of patient-handling techniques).

All procedures involving blood or other potentially infectious material are performed in such a manner as to minimize splashing, spraying, spattering, and generation of droplets of these substances.

Employees who have the potential for being exposed to bloodborne pathogens are asked to participate in the identification, evaluation, and selection of effective engineering and work practice controls. Employees may provide to the dentist or office manager information on an engineering and/or work practice control, and request that it be evaluated.

- *Preventing Contact with Infectious Materials*

Exposure to infectious body fluids and transmission of disease can occur through **direct or indirect contact** with an infected person. Examples of dangers to avoid in the office are:

1. Putting bare hands into the mouth of a patient.
2. Touching the blood or saliva from a patient.
3. Being splashed with blood or saliva that enters through the mucous membranes of the eyes, nose or mouth.
4. Touching contaminated instruments, equipment, work surfaces, or waste.
5. Receiving a cut or puncture wound from a sharp instrument or needle.

6. Touching impressions, or other objects that have been in the patient's mouth.
7. Having contact with airborne microorganisms in splatter or aerosolized debris.

Fluids likely to be encountered in the dental office that are recognized as potentially linked to the transmission of HIV, HBV, or HCV, and to which standard precautions apply, include blood, saliva and gingival fluids. Neither HBV, HCV nor HIV can be transmitted by casual contact in the workplace.

- *Accident Avoidance*

When procedures are not done carefully or correctly, or are done in haste, accidents do happen. Attention should be focused and fatigue avoided to prevent stabs or scrapes from:

1. Burs left in the handpiece, sitting upright in the bracket holder.
2. Removing Metal crowns.
3. Laboratory knives and burs.
4. Scalers, blades, needles or other sharp instruments on the treatment tray.
5. Ultrasonic tips which are exposed in the field of operation.
6. Transport of instruments from the operatory to the instrument sterilization area.

Dental Sharps

1. Needles for anesthetic, needles for periodontal irrigation, surgical blades, burs
2. Dental procedure(s) for which sharp is used: all chairside and laboratory procedures may involve sharps.

Cal/OSHA prohibits the following practices with regards to sharps management:

1. Shearing or breaking of contaminated needles, blades, and other contaminated sharps is prohibited.
2. Contaminated sharps shall not be bent, recapped, or removed from devices.
Exception: Contaminated sharps may be bent, recapped or removed from devices if the procedure is performed using a mechanical device or a one-handed technique, and the employer can demonstrate that no alternative is feasible or that such action is required by a specific medical or dental procedure.
3. Sharps that are contaminated with blood or other potentially infectious material shall not be stored or processed in a manner that requires employees to reach by hand into the containers where these sharps have been placed.
4. Disposable sharps shall not be reused.
5. Do not pick up broken glass directly with your hands. Use mechanical means, such as brush and dustpan or tongs.
6. The contents of reusable sharps containers shall not be accessed unless properly reprocessed or decontaminated.
7. Sharps containers shall not be opened, emptied, or cleaned manually or in any other manner that would expose employees to the risk of sharps injury.

Employees shall:

1. Not use two hands to recap needles.
2. Not put hands into containers that contain reusable sharps, such as curettes or blades, which are being stored prior to decontamination.
3. Assemble and disassemble handpieces in a safe manner and remove burrs using a gauze square and gloves or other technique to prevent puncture wounds.
4. Use effective patient-handling techniques and other methods to minimize the risk of a sharps injury during procedures involving the use of a sharp on a patient.

Procedures in this office to safely manage sharps include:

1. Needles are recapped with a one-handed technique.
2. Recapped needles are placed on the instrument tray and carried to the central sterilization room where the capped needle will be discarded into the sharps container.
3. An instrument is used to retract tissue during anesthetic injections.
4. Sutures and blades are separated from the rest of the surgical set-up to ensure that they are very visible.
5. Instrument trays are transported to and from operatories and central sterilization area in a manner that prevents the accidental dropping of instruments.
6. Brush-scrubbing of debris from contaminated instruments is only permitted after they are pre-cleaned in an ultrasonic cleaner. Gloves are to be worn when brush-scrubbing.

Management of Waste*Medical waste includes the following:*

1. Contaminated sharps, (e.g., needles, disposable syringes, blades, endo files, burs, temporary crowns with sharp edges, and carpules with aspirated blood).
2. Disposables that “drip blood” or other body fluids when compressed or flake dried blood when shaken (e.g., dressings, gauze, cotton rolls).
3. Fluid blood.
4. Human tissues.
5. Teeth that are deemed ***infectious by the attending medical/dental professional***.
6. Waste from a person infected with diseases which are highly communicable to humans, (CDC Biosafety Level IV organisms, **see Medical Waste Disposal Plan**).
7. Pharmaceutical waste.

Personal protective equipment:

Gloves and gown as necessary, shall be worn while handling medical waste in the office to prevent employee exposure to body fluids.

Containment Requirements for Contaminated Sharps:

Rigid, Puncture resistant, Leak-proof on sides and bottom, labeled “Sharps Waste,” easily accessible to personnel and located as close as is feasible to the immediate area where sharps are used or can be reasonably anticipated to be found, Maintained

upright, Replaced as necessary to avoid overfilling, Portable, Closed prior to moving or transport.

Non-regulated minimally-contaminated waste is:

1. Placed into containers that are lined with unlabeled plastic bags. The plastic liner bags can be disposed of as solid waste.
2. Dropped through holes in the counters into unlabeled plastic-lined containers.

No specimens of blood or other potentially infectious materials are handled in this office.

Cleaning, Decontamination and Hygiene

The housekeeping procedures insure that all areas of the office are clean and sanitary. The methods for cleaning and decontamination shall be effective and appropriate for the location within the office, type of surface or equipment to be cleaned and disinfected, type of contamination, and tasks or procedures performed in the area. Included in the housekeeping procedures:

1. Patient treatment areas and work surfaces and equipment must be cleaned and decontaminated with an appropriate disinfectant between patients.
2. Overtly contaminated work surfaces and spills of body fluids shall be cleaned and decontaminated immediately or as soon as possible.
3. Following the initial cleaning, disinfection shall be done with one of the chemical germicides that are approved for use as a hospital disinfectant and is tuberculocidal when used in recommended dilutions.
4. Reusable receptacles shall be inspected regularly, and cleaned and decontaminated when visibly contaminated.
5. If used, protective coverings, such as plastic wraps, shall be removed and replaced between patients.
6. Sweeping, handling solid or liquid wastes, shall be done without allowing personal contamination.

Personal Items and Contamination

1. No eating, drinking, applying cosmetics, handling contact lenses or smoking in work areas.
2. No food or drink in refrigerators, on countertops, or on shelves where blood or other potentially infectious materials are present. Food and drink may be kept in break room.

Minimizing Surface Contamination

1. The office personnel should attempt to minimize splash, spray and splatter through use of such things as: high velocity evacuation, dry shield
2. Minimize the number of items that are contaminated during care by:
 - pre-dispensing medicaments and armamentaria
 - use of over-gloves or barriers over contaminated treatment gloves to obtain clean supplies
 - use of disposable barriers on equipment and surfaces

Hygiene

Staff are instructed to wash their hands with soap and running water, and may use alcohol handrubs as recommended by CDC if hands are not visibly soiled:

1. When gloves become torn or defective.
2. Immediately or as soon as feasible after removal of gloves or other personal protective equipment.
3. After contact with blood or other potentially infectious material with bare hands or any other skin.
4. Hand washing facilities are available in the following areas of the office: hallway sinks, bathroom and breakroom
5. Alcohol-based handrub dispensers are located throughout the office.
6. Mucous membranes shall be flushed with water immediately or as soon as feasible after contact with blood or other potentially infectious material.

Personal Protective Equipment

1. To prevent exposure to body fluids, ***personal protective equipment must be used that does not permit potentially infectious materials to reach employee's work clothes, street clothes, skin, eyes, nose and mouth.*** In this office, PPE will be used during all clinical procedures, radiographic procedures, cleaning, disinfecting and sterilization procedures, lab procedures and trash disposal.
2. *Gloves, masks, faceshields, goggles, gowns, scrub jackets are available.*
3. Remove personal protective equipment before leaving work area (e.g., before entering uncontaminated areas such as the lounge or reception area.).
4. In situations where gross contamination is reasonably anticipated, use shoe covers and hoods or caps (e.g., handpiece use during major oral surgery in an operating room situation).
5. Immediately replace garment penetrated by blood.
6. Replace garments that are wet or visibly contaminated with blood after patient treatment or as soon as possible.
7. After removal, contaminated linens that are to be laundered shall be placed into the washing machine or a container located near washing machine.
8. Wear gloves in situations of potential exposure.
9. Replace contaminated gloves as soon as possible.

Employer's Responsibility on Personal Protective Equipment

1. Provide at no cost to employees.

Hepatitis B Vaccination and Bloodborne Pathogen Post-Exposure Evaluation and Follow-Up

1. Hepatitis B Vaccinations for Potentially Exposed Employees

The risk of transmission of Hepatitis B is known to be a serious health risk for dental personnel. The HBV vaccine is an effective preventative measure that is strongly recommended. All employees likely to be exposed to infectious fluids are instructed about the hazards of contracting Hepatitis B and are advised to have the Hepatitis B vaccination at employer's cost. The vaccine is not necessary if the employee tests positive for HBV immunity, if the vaccine is contraindicated for a medical condition, or if the employee has already received it.

Refusal to have the vaccine can be reversed at any time. Refusal of the vaccine and records of vaccine administration are kept in accordance with Cal/OSHA regulations.

2. Reportable Exposure Incidents

Exposure incidents that must be reported and documented include eye, mouth or other mucous membrane, non-intact skin or parenteral contact with blood, saliva or other potentially infectious materials. Incident reports will be kept in accordance with Cal/OSHA regulations, including routes and circumstances of exposure. Employees who have an exposure are required to report the exposure to Dr. Leong immediately and must be given instruction in follow-up procedures. This dental office will provide for post-exposure evaluation and follow-up as required by Cal/OSHA.

Follow-Up Procedures

In the event of exposure incident:

1. Exposed employee should wash skin with soap and water and flush mucous membranes.
2. Exposed employee must report incident immediately to Dr. Leong.
3. The exposed employee is referred as soon as possible to a health care provider who will follow the current recommendations of the U. S. Public Health Service, Centers for Disease Control and Prevention recommendations for testing, medical examination, prophylaxis and counseling procedures.
4. The U.S. Public Health Service currently recommends post-exposure prophylaxis for HIV be started within 1 to 24 hours of an exposure incident and notes that use of chemoprophylaxis is a clinical decision dependent on the characteristics of the injury. Certain information must be provided to the health care provider, and this is detailed further in this section.

5. The exposed employee may refuse any medical evaluation, test or follow-up recommendation. This refusal is documented.
6. The source individual is identified (if possible and if permitted by law) and, with the source individual's consent, this dental office provides for testing the individual for HBV, HCV and HIV carrier status.

The health care provider will follow these procedures:

1. With the employee's consent, the health care provider will collect the employee's blood to establish HBV, HCV and HIV status. If the blood is collected and the employee declines testing, the blood must be stored for at least 90 days in case the employee consents to testing later.
2. Inform the employee of all test results, including tests on the source individual if such testing was done.
3. Counsel employee concerning infectious status, test interpretation and need for post-exposure prophylaxis.
4. Prescribe prophylactic measures as medically indicated and as recommended by the U. S. Public Health Service.
5. Evaluate any reported illnesses to determine any relation to HBV, HCV or HIV infection.
6. Provide a written opinion, as required by Cal/OSHA, to this dental office.

Health Care Provider's Written Opinion

The health care provider's written opinion on the exposure incident must be provided to the employee within 15 days of completion of the evaluation. All other findings remain confidential and are not included in the written report.

Instrument Sterilization Protocol

Personal Protective Equipment

- All DHCP shall wear surgical facemasks in combination with either chin length plastic face shields or protective eyewear whenever there is potential for aerosol spray, splashing or spattering of the following: droplet nuclei, blood, chemical or germicidal agents or OPIM. Chemical-resistant utility gloves and appropriate, task specific PPE shall be worn when handling hazardous chemicals.
- Protective attire shall be worn for disinfection, sterilization, and housekeeping procedures involving the use of germicides or handling contaminated items. All DHCP shall wear reusable or disposable protective attire whenever there is a potential for aerosol spray, splashing or spattering of blood, OPIM, or chemicals and germicidal agents.
- When processing contaminated sharp instruments, needles and devices, DHCP shall wear heavy-duty utility gloves to prevent puncture wounds.

In the Operatory to the Sterilization Area

- Single use disposable items such as prophylaxis angles, prophylaxis cups and brushes, tips for high-speed evacuators, saliva ejectors, air/water syringe tips, and gloves shall be used for one patient only and discarded.
- Needles shall be recapped only by using the scoop technique or a protective device. Needles shall not be bent or broken for the purpose of disposal. Disposable needles, syringes, scalpel blades, or other sharp items and instruments shall be placed into sharps containers for disposal as close as possible to the point of use according to all applicable local, state, and federal regulations.

Preparing the Instruments/Items for Sterilization or Disinfection

- All germicides must be used in accordance with intended use and label instructions.
- Cleaning must precede any disinfection or sterilization process. Products used to clean items or surfaces prior to disinfection procedures shall be used according to all label instructions.

Sterilization and Disinfection

- Critical instruments, items and devices shall be discarded or pre-cleaned, packaged or wrapped and sterilized after each use. Methods of sterilization shall include steam under pressure (autoclaving), chemical vapor, and dry heat. If a critical item is heat-sensitive, it shall, at a minimum, be processed with high-level disinfection and packaged or wrapped upon completion of the disinfection process.
- Semi-critical instruments, items, and devices shall be pre-cleaned, packaged or wrapped and sterilized after each use. Methods of sterilization include steam under pressure (autoclaving), chemical vapor and dry heat. If a semi-critical item is heat sensitive, it shall, at minimum, be processed with high level disinfection and packaged or wrapped upon completion of the disinfection process.
- All high-speed dental hand pieces, low-speed hand pieces, rotary components, and dental unit attachments such as reusable air/water syringe tips and ultrasonic scaler tips, shall be packaged, labeled and heat-sterilized in a manner consistent with the same sterilization practices as a semi-critical item.
- Non-critical surfaces and patient care items shall be cleaned and disinfected with a California Environmental Protection Agency (Cal/EPA)-registered hospital disinfectant (low-level disinfectant) labeled effective against HBV and HIV. When the item is visibly contaminated with blood or OPIM, a Cal/EPA-registered hospital intermediate-level disinfectant with a tuberculocidal claim shall be used.
- All intraoral items such as impressions, bite registrations, prosthetic and orthodontic appliances shall be cleaned and disinfected with an intermediate-level disinfectant before manipulation in the laboratory and before

placement in the patient's mouth. Such items shall be thoroughly rinsed prior to placement in the patient's mouth.

- Devices used to polish, trim, or adjust contaminated intraoral devices shall be disinfected or sterilized, properly packaged or wrapped and labeled with the date and the specific sterilizer used if more than one sterilizer is utilized in the facility.
- Proper functioning of the sterilization cycle of all sterilization devices shall be verified at least weekly through the use of a biological indicator (such as a spore test). Test results shall be documented and maintained for 12 months.

Instrument Storage

These packages or containers shall remain sealed and shall be stored in a manner so as to prevent contamination, and shall be labeled with the date of sterilization and the specific sterilizer used if more than one sterilizer is utilized in the facility.

If packaging is compromised, the instruments shall be recleaned, packaged in a new wrap, and sterilized again. Sterilized items will be stored in a manner so as to prevent contamination.

Sharps waste

This dental practice generates sharps waste. .

Sharps waste is placed in containers easily accessible and located in the sterilization room.

Sharp containers used in this dental practice are:

- Cleared by FDA as sharps containers.
- Not overfilled and maintained upright.
- When ready for disposal, taped closed or lid tightly closed.
- Labeled with the words "Biohazard" and the international biohazard symbol, or "Sharps Waste."
- Picked up by a CDPH registered waste hauler or shipped out of the office

Handling of Sharps Waste: Handle all sharps in a manner to prevent accidental punctures and personal contamination. Needles should never be bent or cut during disposal. Never put hands into a container for contaminated sharps.

Employee Injury Protocol

All employee injuries must be reported immediately to Dr. Leong.

Injury Type	Protocol
Exposure to another person's blood or saliva	Follow the Bloodborne Pathogens Post Exposure Management Protocol
Eyes exposed to chemical	Use eyewash; injured employee should seek medical evaluation
Minor scrapes, cuts or burns	Wash injured area and apply first aid
Non-life-threatening	Seek medical assistance at a physician's office or medical clinic
Severe, life-threatening	Call 911